

Silicon platforms help to enable vertical integration by bringing high-performance digital, analogue, memory and radio frequency (RF) technologies—all of which have historically been separate on a single chip. The design of such a complex chip is compounded by the intricacies of designing on an advanced processing node like 28nm or below.

Fig. 2 shows a platform example along with the critical steps needed to pull the different technologies together effectively. There are multiple ARM cores, a 3D graphics processing engine, audio and video digital signal processors, multimedia and image processing, USB and other inputs/outputs, and complex memory subsystem. Such a platform is likely to find itself in any typical vertically-integrated mobile or consumer device. Particularly important in this flow are full-chip mixed-signal simulation, IP characterisation, digital sign-off, and analogue and digital physical design—all on an integrated data model.

Historically, analogue and digital verification have relied on different approaches. Huge swaths of digital circuitry can be verified by logic simulation; relatively smaller analogue blocks have been verified by simulation program with integrated circuit emphasis (SPICE). On the same platform, full-chip simulation must combine the capacity for the whole chip consisting of millions of transistors, while still maintaining SPICE-level accuracy. A fast multicore simulator is typically required to manage these extreme requirements.

Meanwhile, any IP being integrated into the platform must be characterised in order to reduce design pessimism while still ensuring that it will play well together with the other components on the chip. This again requires multicore circuit simulation in combination with efficient standard cell, inputs/outputs and memory models.

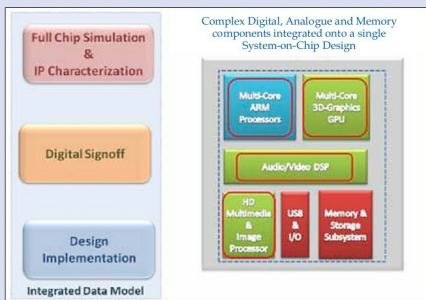


Fig. 2: Fully integrated solutions for emerging silicon

Digital sign-off exercises the logic through hundreds of operational modes and all of the various environmental corners that the chip might find itself operating in. This means corralling numerous timing, parasitic extraction and physical verification scenarios while managing the computing resources needed to process it all. These different aspects need to be verified concurrently in order to get through the entire suite in time to hit the market window. All pieces must work together with high performance and at the same time demonstrate extremely high accuracy in order to correlate well with low-level SPICE and 3D solvers.

When it comes to physical design implementation, the required design capacity is ballooning rapidly. Along with this growth, a single tool must be able to deal with clock tree design, multiple scenarios for routing and timing-driven placement, routing congestion, analogue shape-based routing and memory routing. State-of-the-art solutions will need to be able to handle sophisticated power-management techniques from dynamic voltage/frequency scaling (DVFS) to multiple power islands to

clock gating. On top of all of this, in order to reduce the number of design iterations it is increasingly important that sign-off quality engines are used during implementation. That reduces the chances of last-minute surprises when going for tape-out.

Given integrated tools that can manage the intricacies of assembling a single chip with such different technologies, vertically-integrated companies can develop robust platforms that will carry their products through the critical first-growth stage. It is here that the company positions itself for future sustained success.

Whether gunning for the exploding mobile market, emerging cloud computing market or the giant consumer market, dedicated platforms are where the critical trade-offs between power, performance, storage, connectivity and all other critical parameters of a system take place. Each market demands its own mix of characteristics, meaning that tailored platforms must be brought to market quickly and efficiently. In the end, it's the silicon design tools that make this possible, and the onus is on tool providers to enable rapid success in vertically-integrated markets. ●